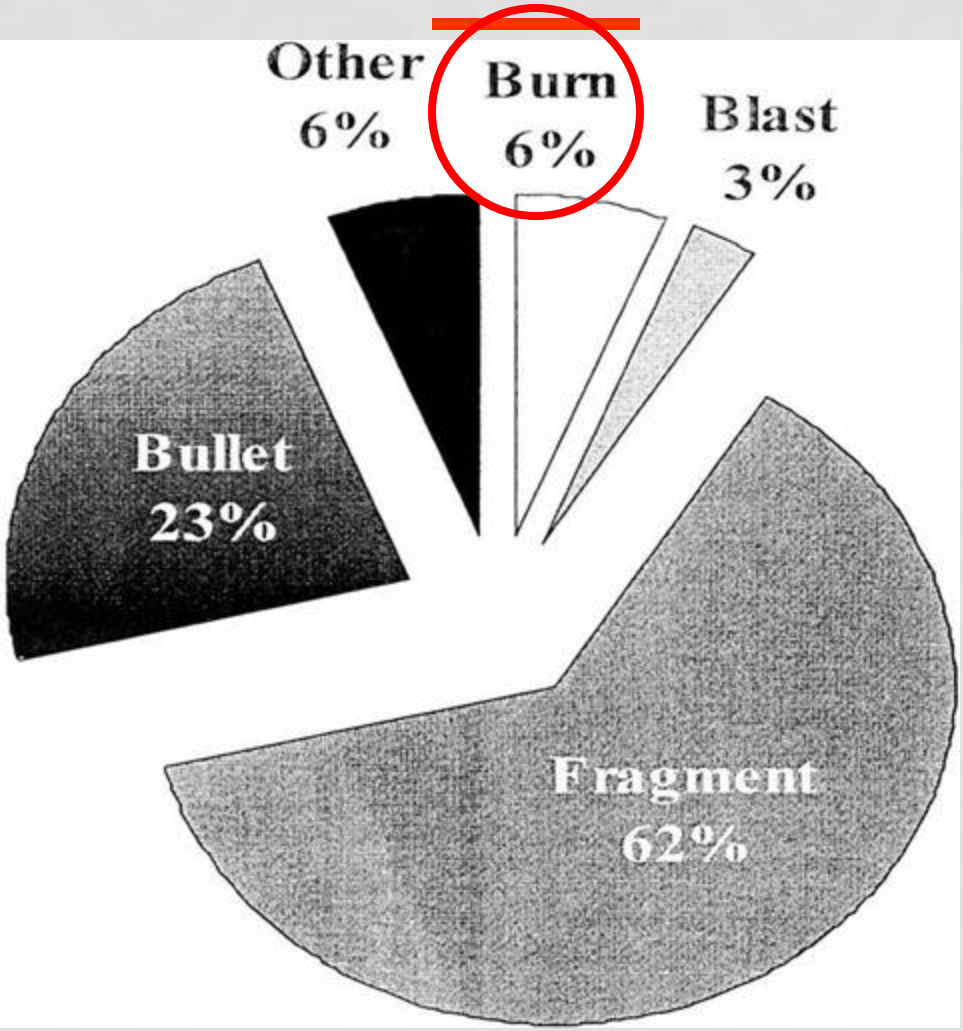
BURNS

Anthony Papp, MD, PhD
Medical Director, Burn Unit, VGH
Clinical Professor, UBC

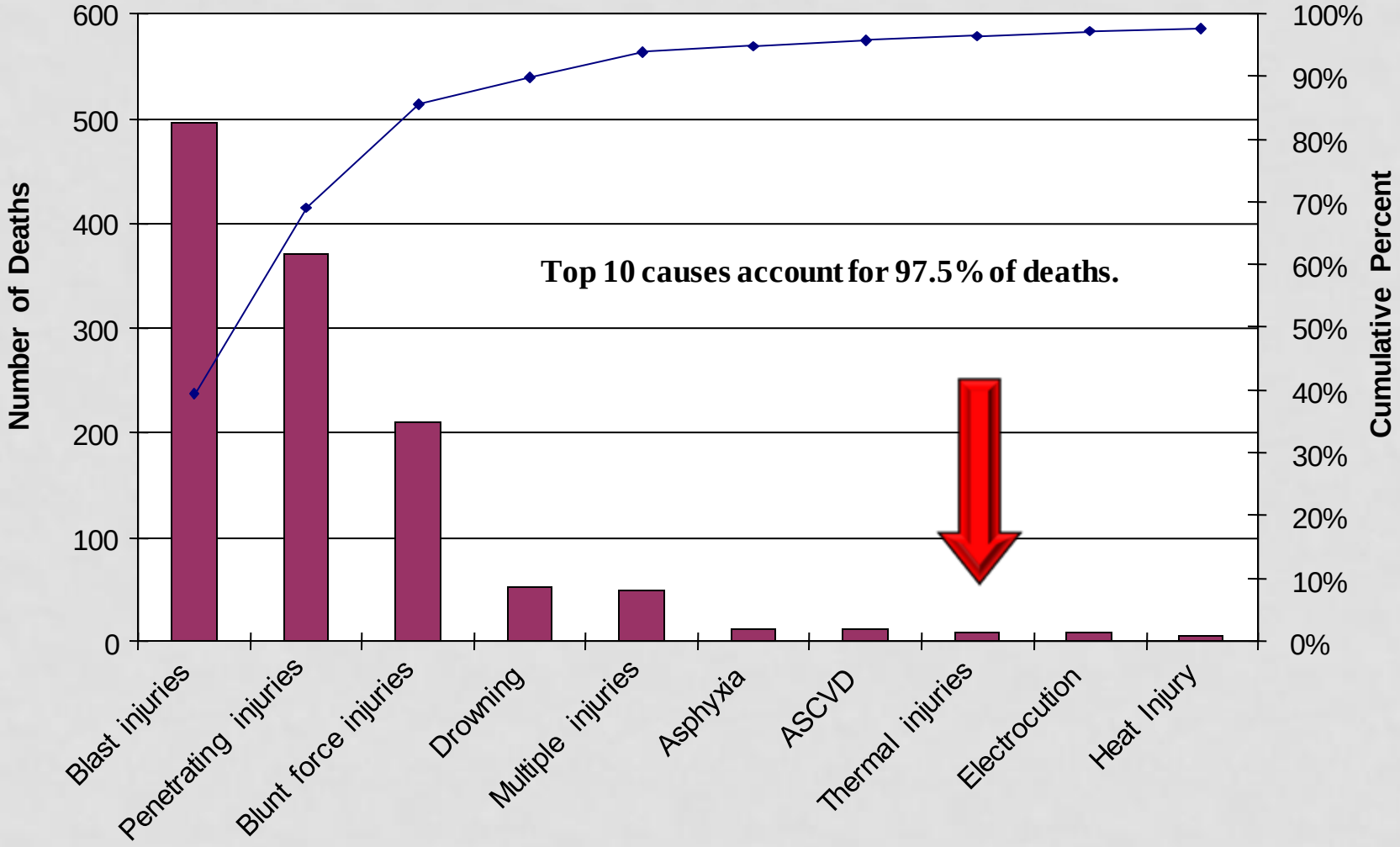
Mechanism of Wounding – WDMET (wound data and munitions effectiveness team) Database Vietnam 1967-1969 [1]



1. Bellamy RF. Textbook of Military Medicine, Anesthesia, and Perioperative Care of Combat Casualty. 1995

TOP 10 CAUSES OF OIF DEATHS

3/19/2003 – 11/30/2004



Wartime Care at Sea of Patients with Severe Thermal Injury.

Buescher TM, Cancio LC, Holcomb JB.

J Burn Care and Rehabilitation 2004;25:S75(abstract)

N = 6 requiring surgery for burns

mean age 24 yrs

mean TBSA 47% (27-69)

- **2/6 invasive burn wound infection**
- **1/6 mesenteric ischemia**
- **1/6 acalculous cholecystitis**
- **1/6 suppurative thrombophlebitis**
- **4/6 pneumonia**
- **6/6 bacteremia**

THE PATIENT WITH BURNS IS A TRAUMA PATIENT!

Resuscitation

- ✓ATLS Protocol
- ✓Ensure adequate tissue perfusion
- ✓Prevent further injury

GENERAL PRINCIPLES

- Patient history
- Primary survey
- Resuscitation
- Secondary survey

HISTORY

Mechanism of injury

- Flame, scald, flash, contact, electrical, chemical, tar…

When did injury occur?

- Time of burn is starting point of resuscitation!

Where and how did injury occur?

- Transportation time, delay in fluid resuscitation

Is there a potential for inhalation injury?

- Enclosed space?

PATIENT HISTORY

Patient characteristics

- Age
- Co-morbid conditions

Was there other mechanical trauma coincident with the burn?

- Explosion, fall, MVC

PRIMARY SURVEY ABC'S

Airway

- ✓Anticipate need for early intubation
- ✓Smoke and noxious products of combustion
- ✓Look for erythema and edema of airway

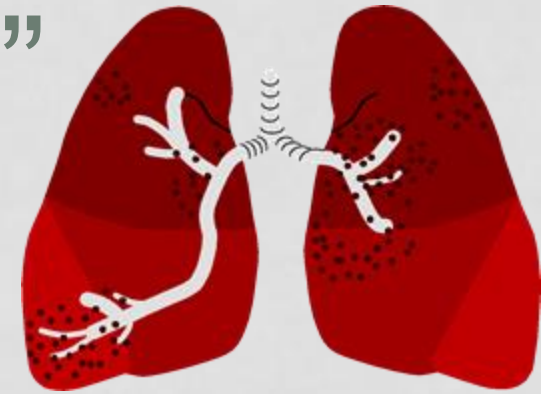
Evaluate and Secure Airway

What are some clinical signs of inhalation injury?

CLINICAL SIGNS OF INHALATION INJURY

- Stridor
- Hoarseness
- Facial burns
- Intraoral burns
- Singed nasal hairs
- Carbonaceous sputum
- Impaired level of consciousness

“BREATHING”



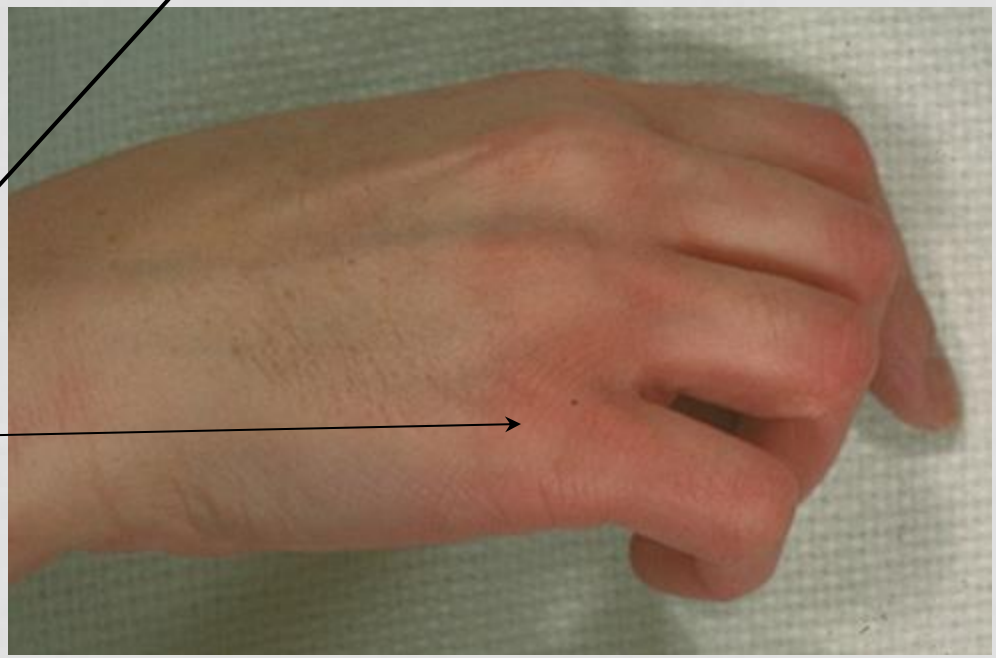
- ✓ Chemical burns caused by smoke inhalation
- ✓ Affect lower airway and alveoli
- ✓ Progressive respiratory compromise
  - ➢ Can occur up to 72 hours post-injury

When should you suspect Carbon Monoxide Poisoning?

What is the “classic” physical finding?

CO POISONING

- History of smoke exposure in an enclosed space
- A finding of disproportionate obtundation
- Cherry-red skin discoloration



WAS THE PATIENT TRAPPED IN A CLOSED SPACE?

If so, suspect carbon monoxide (CO) poisoning

- ✓Measure carboxyhemoglobin levels
- ✓CO: affinity for Hb is 200x that of 0_2

➢ **TX: 100% Oxygen**

“CIRCULATION”

- ✓ Pulses
- ✓ Loss of skin = massive loss of salt and fluid
- ✓ Monitor volume replacement and tissue perfusion

“DISABILITY”

- ✓ Look for mechanical trauma
- ✓ Altered mental status
  - ➢ GCS (3-15)
  - ➢ Hypoxia and hypovolemia
- ✓ Neuro deficits
  - ➢ Closed head injury

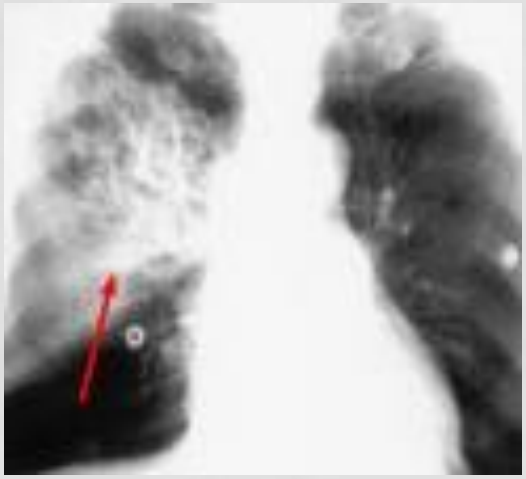
“EXPOSURE”

- ✓ Remove clothing
  - ➢ Heat
  - ➢ Chemicals
- ✓ Lightly brush away dry powders
- ✓ Copiously irrigate skin (chemicals)
- ✓ Remove potential “tourniquets”
- ✓ Keep the patient warm during and after exposure
  - ➢ Room temperature
  - ➢ Warm blankets
  - ➢ Warm fluids etc

RESUSCITATION

- ✓ Each problem uncovered
  - ➢ Corrected in sequence
- ✓ Intubation
  - ➢ Treat like any other trauma patient
  - ➢ Recognize potential for airway compromise
- ✓ Bronchoscopy
  - ➢ Important for evaluation and **diagnosis** of upper airway injury / inhalation injury
  - ➢ Fast diagnosis, reasonably easy to perform
  - ➢ Gives a good idea of the status of the airways

RESUSCITATION (X-RAYS)



Chest X-ray

- Alveolar injury lags behind clinical picture
- Better for later suspicion of pneumonia/ARDS

C-spine, multitrauma patients/falls/explosion

- Once cleared, elevate HOB 30^0

RESUSCITATION

Vascular access

- ✓Large bore peripheral IV's
  - ➢Use non burned tissue
- ✓Central lines
  - ➢Use non-burned tissue if possible

If no burn-free site

- Place through burn eschar
- May be excised, if needed

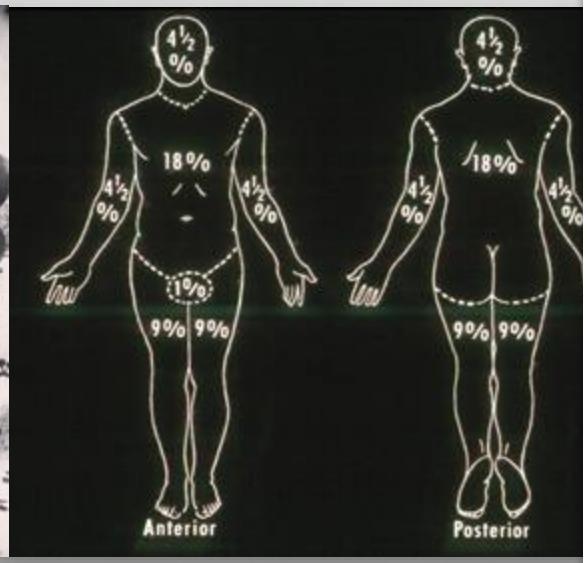
SECONDARY SURVEY

- ✓ Coincident trauma diagnosed and treated
- ✓ Burns are further characterized
  - ➢ Determination of <u>extent</u> and <u>depth</u>

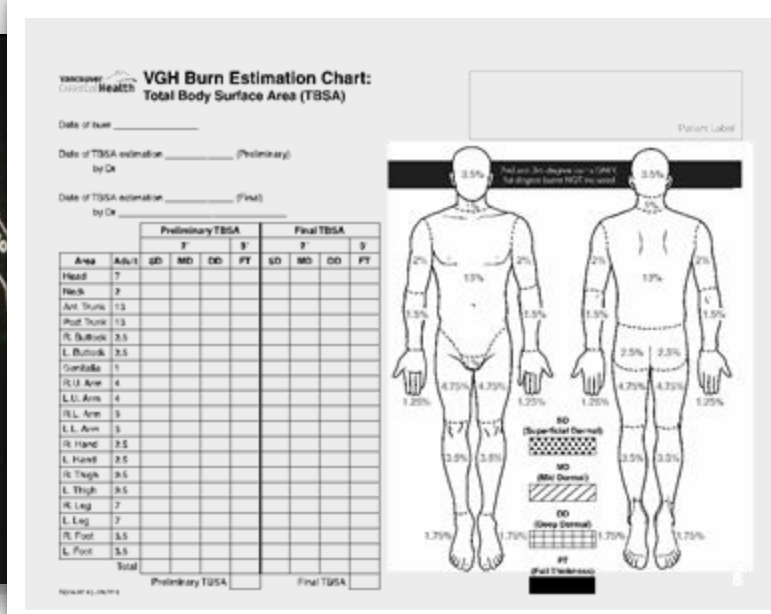
BURN EXTENT, TBSA

Palm method

Rule of nines



Lund Browder nomogram



VGH Burn Estimation Chart:
Total Body Surface Area (TBSA)

Date of burn _____

Date of TBSA estimation _____ (Preliminary)
by Dr

Date of TBSA estimation _____ (Final)
by Dr _____

Area	Adult	Preliminary TBSA 3° SD	MD	DD	FT	Final TBSA 3° SD	MD	DD	FT
Head	7								
Neck	2								
Ant. Trunk	13								
Post. Trunk	13								
R. Buttock	2.5								
L. Buttock	2.5								
Genitalia	1								
R.U. Arm	4								
L.U. Arm	4								
R.L. Arm	3								
L.L. Arm	3								
R. Hand	2.5								
L. Hand	2.5								
R. Thigh	5.5								
L. Thigh	5.5								
R. Leg	7								
L. Leg	7								
R. Foot	3.5								
L. Foot	3.5								
Total									
		Preliminary TBSA				Final TBSA			

ESTIMATING BURN AREA

Adding it all up

TBSA burned

- ✓ Sum of the second *plus* third-degree burns*
- ✓ First-degree burns are mapped
  - ➢ Not included in this calculation

*This is amount used to calculate fluid requirements

RESUSCITATION

Effective fluid resuscitation in the first 48 hours post burn injury is critical in preventing end organ hypoperfusion and improves survival

Parkland formula (4ml/kg/TBSA) tends to over-resuscitate patients

We use 3ml/kg/TBSA

- ✓ Current ABLS guideline is to start with 2 ml/kg/TBSA

- Fluid calculation is an ESTIMATE
- Individual patient response dictates therapy
- ↑ fluid needs common in patients with
  - Associated injuries
  - Electric injury
  - Inhalation injury
  - Resuscitation delay
  - Prior dehydration
  - ETOH &/or drug abuse
  - Very deep burn injury
  - Small children with small burns

QUESTIONS

What do we use for fluid?
- ✓ Ringers lactate

Urine output goal?
- ✓ 30-50 ml/kg regardless of body weight

Boluses, yes or no?
- ✓ No! You increase or decrease fluids by 20% if UO low/high 2 consecutive hours

Colloids/albumin?
- ✓ Yes, as soon as 8-12h after injury if required

MAP goal?
- ✓ No need to be 65mmHg.
- ✓ >55mmHg is fine as long as UO>30ml/h, lact <4mmol/l and ScvO2>70%

Norepinephrine?
- ✓ Ideally not. Why?

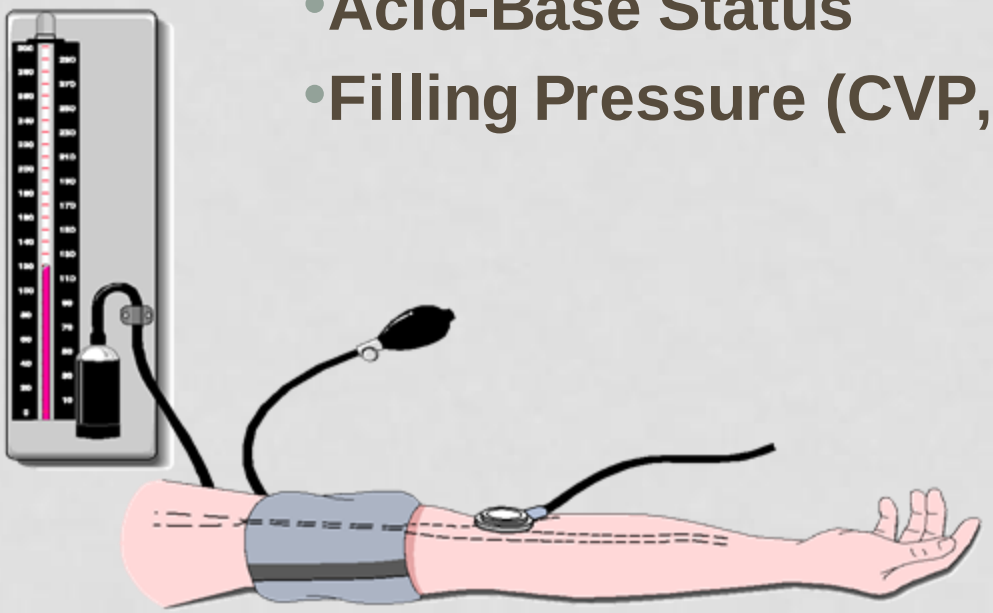
REMEMBER!

The timing starts when the burn occurred

***<u>Not</u>* when the patient arrived in the treatment facility**

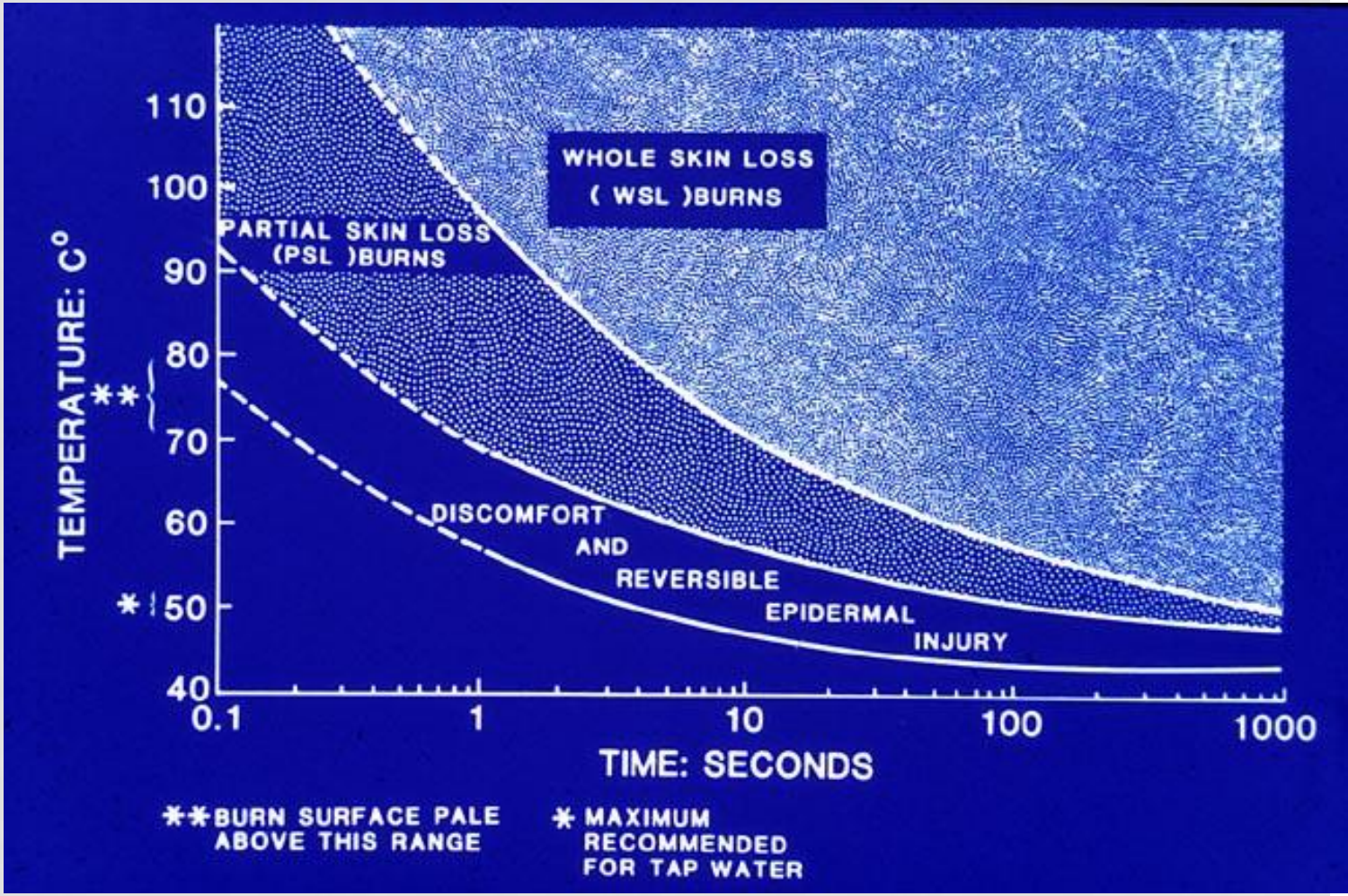
PARAMETERS FOR FLUID RESUSCITATION

- **Hourly Urine Output**
- **Heart Rate, Blood Pressure**
- **Hematocrit**
- **Acid-Base Status**
- **Filling Pressure (CVP, PAWP)**

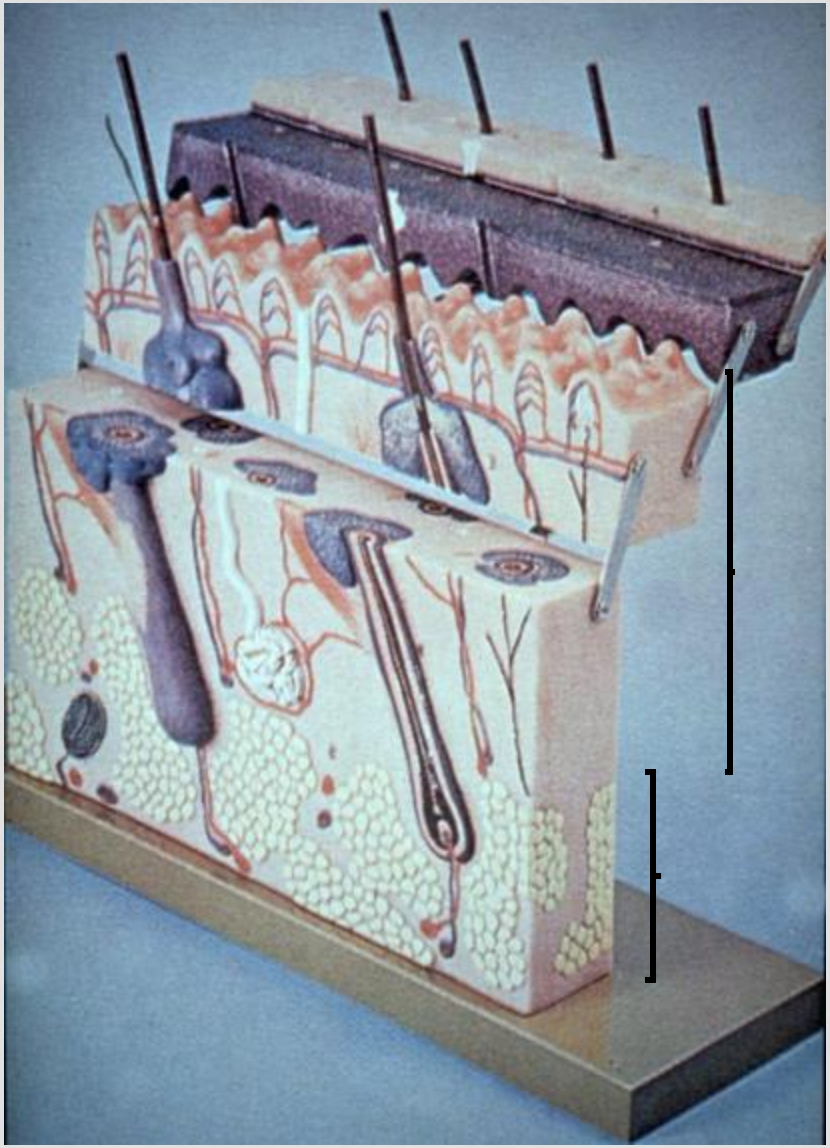


BURN DEPTH

Contact time
Temperatur
e



Classification of Burn Depth



1 degree

2 degree
superficial
mid dermal
deep dermal

3 degree

FIRST DEGREE BURN

- ✓Superficial (sunburn)
- ✓Erythema, pain, absence of blisters
- ✓Consists of epidermal damage alone
- ✓Do not include in TBSA estimation

Heals within 3 to 6 days

- ➢No scarring

SECOND DEGREE BURN

Involves

✓Entire epidermal layer

✓Different parts of underlying dermis

✓Mottled and red, painful, swelling and blisters

- Superficial dermal burns
  - Usually quite painful
  - Erythemetous with blebs and bullae
  - Even air motion across skin hurts
  - Heal within 2 weeks

- Mid-dermal burns
  - Can be treated either conservatively or operatively
  - Heal within 3-6 weeks

- Deep partial-thickness burns:
  - Sensation impaired to a variable degree
  - Need operative treatment

THE HAIR FOLLICLE

KEY TO HEALING OF THE PARTIAL-THICKNESS BURN AND DONOR SITE WOUND

Epithelialization

THIRD-DEGREE

- ✓Destruction of all epidermal and dermal elements
- ✓Burn into subcutaneous fat or deeper
- ✓Skin is charred and leathery
- ✓Pearly-white sheen / waxy
- ✓Generally not painful (nerve endings are dead)

Deep burn

White

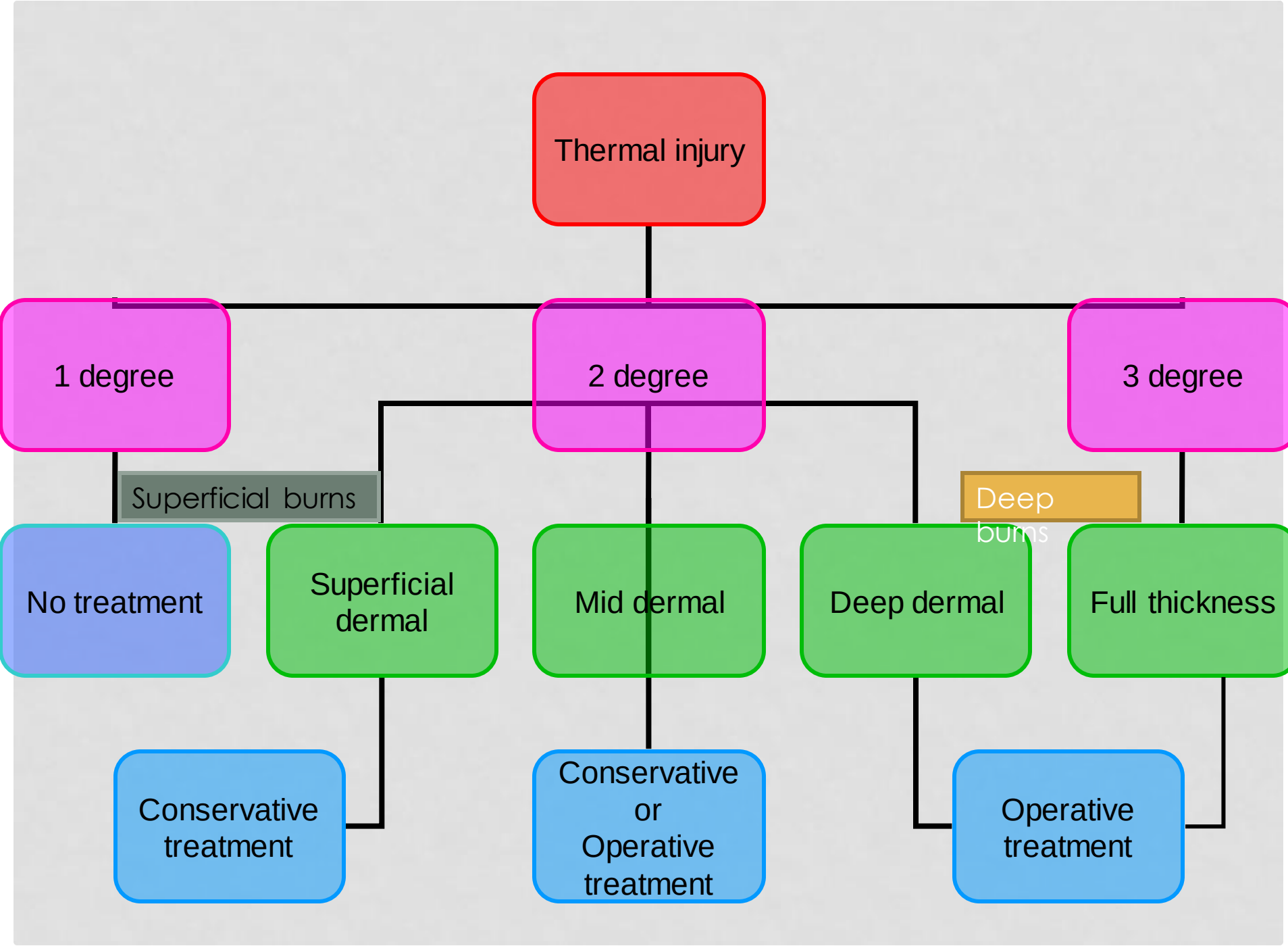
Black

Several colors

Yellow, green, brown

FOURTH-DEGREE

- ✓Full-thickness
- ✓Extending into muscle, tendons or bones
- ✓Typically involves appendage
- ✓Black and dry
- ✓No pain


Thermal injury
1 degree
2 degree
3 degree
Superficial burns
Deep burns
No treatment
Superficial dermal
Mid dermal
Deep dermal
Full thickness
Conservative treatment
Conservative or Operative treatment
Operative treatment

Burn depth progression

FIELD HOSPITAL

- ✓ Little care is required for burn wounds prior to transfer
- ✓ If gross contamination:
  - ➢ Gentle washing and debridement
- ✓ If patient cannot be rapidly transferred
  - ➢ Apply antibiotic ointments
  - ➢ Occlusive dressings
  - ➢ Helps prevent evaporative heat loss

INITIAL PHASE OF TREATMENT

- After initial cooling of the burn
  - ➢Keep patient warm during transport
- Cover burn wounds
  - ➢Cover with sterile gauze
  - ➢Clean dry sheet, blankets

TETANUS

Burns are tetanus prone wounds

- Provide tetanus prophylaxis to all burn patients
- But tetanus is rare complication

Determine tetanus status

- If *without* current tetanus immunization
  - Requires tetanus immune globulin (TIG)
  - Requires complete series if Td

ANTIBIOTICS?

<u>Used only for specific infections</u>

Prophylactic antibiotics

- Do not prevent sepsis
- Increase risk of resistant microorganisms

Administer at first sign of clinical suspicion of invasive infection

- Wound infection with fever, leucocytosis, CRP↑

Antibiotics are not used for wound infection per se

- Topical wound care products

What is an escharotomy?

ESCHAROTOMY

- When skin destroyed by heat it looses elasticity
- Burned tissue unable to expand to accommodate edema
- Neurovascular damage begins when interstitial pressure reaches 40mmHg
- P.E: <u>Unreliable</u>
  - Doppler
  - Direct measurement of tissue pressure

ESCHAROTOMY

Circumferential third-degree burns of extremities or digits

- Tourniquet-like pressure
- Compromises blood flow
- Escharotomy releases the tissues
  - Lowers tissue pressures and returns flow

Has to go through the burn both in depth and length

CHEST WALL ESCHAROTOMY

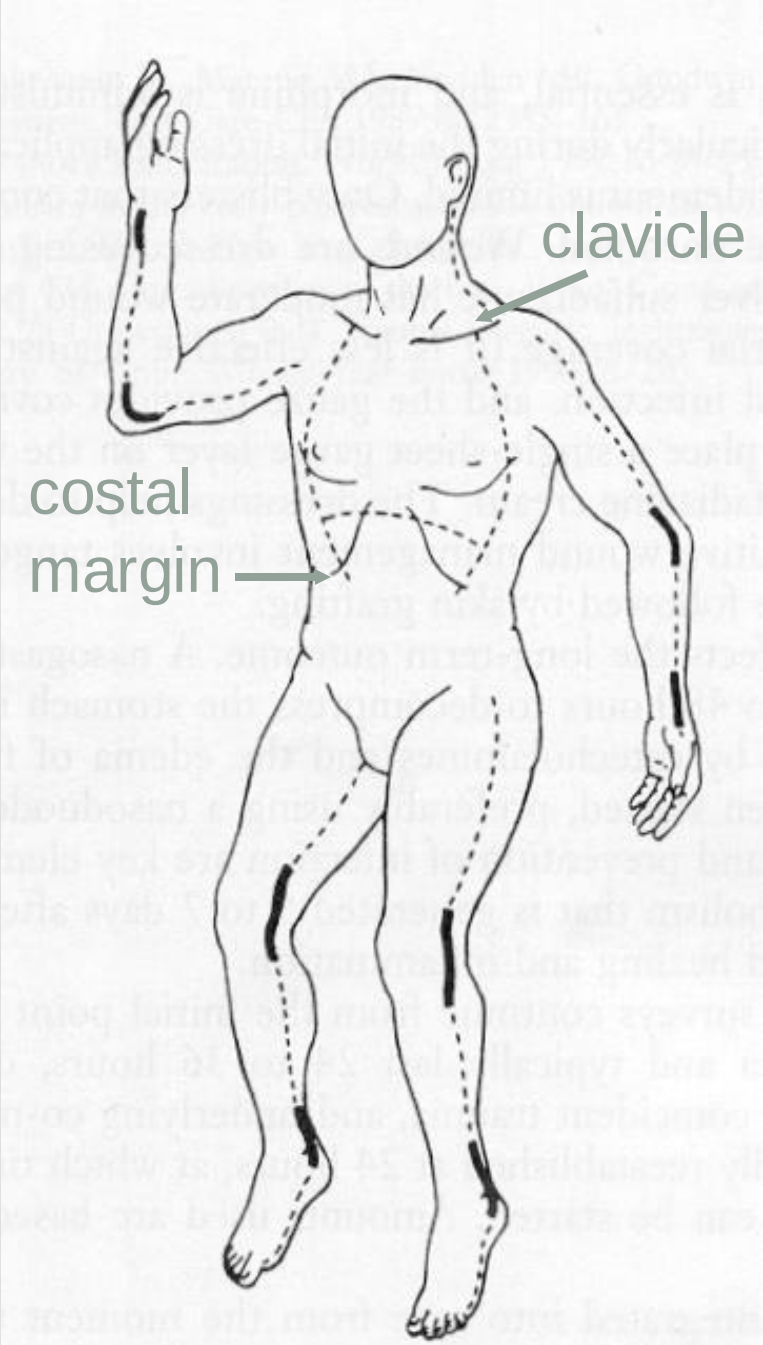
Circumferential deep thermal burns to chest and abdomen

Eschar compromises chest wall compliance

- Signs & Symptoms
  - Deteriorating ABG's
  - Elevated peak airway pressures on ventilator
  - Decrease in urine output
    - Abdominal compartment syndrome

ESCHAROTOMY INCISIONS

- Cut through the eschar along parallel lines
- Incisions through eschar
- Can be done at bedside
- Use electrocautery



FASCIOTOMY

Deep thermal injury

- ✓High voltage electrical injury
- ✓Soft tissue trauma / vascular injury

Escharotomy fails to restore tissue blood flow

Fasciotomy of involved compartments

COMPLICATIONS IN ACUTE BURN MANAGEMENT

Inadequate resuscitation!

Burn wound infection

- ✓ May lead to sepsis and mof
- ✓ Diagnosed by wound biopsy
- ✓ IV antibiotics, debridement

Pulmonary

- ✓ Staph. Aureus and gram neg. pneumonias
- ✓ ARDS

Acute Bacterial Endocarditis

GI

- ✓ stress ulcers, ileus, acalculous cholecystitis, ACS

DEFINITIVE CARE

Regional Burn Center

- Burn Intensive Care Unit
- Wound care
- Wound excision
  - Tangential or fascial excision
- Biologic/other dressings
- Nutrition

Possible Topicals

Previously

- ✓Chlorhexidine
- ✓ Polysporin
- ✓SSD
- ✓ $AgNO^3$
- ✓Sulfamylon

Downsides

- ✓Need for frequent dressing changes
- ✓Takes a lot of nursing time
- ✓Every time a wound dressing is done
- ✓Wound temperature decreases
- ✓Slows down wound healing

The Use of Silver for its Medicinal Properties

- Antiquity 1000 BC Vessels for water
- Ravelin 1869 Antimicrobial effect
- Crede 1881 Ophthalmic rinse
- von Behring 1887 Kills Anthrax
- Physicians 1930's Systemic antimicrobial
- Burke 1964 Silver nitrate for burns
- Fox 1968 Silver Sulfadiazine
- Various 1990's Wound care products

Silver as an Antimicrobial

- **Broad Spectrum**
- **Fast acting**
- **Effective**
- **Low Toxicity**
- **No genetic transmission of resistance**
- **Silver has to be bioavailable**

Not all silver is the same!

Acticoat

nanocrystaline silver

FRESH BURNS

3 dressing changes Q5days

TREATING INFECTION

3 days later

POSTOPERATIVE CARE

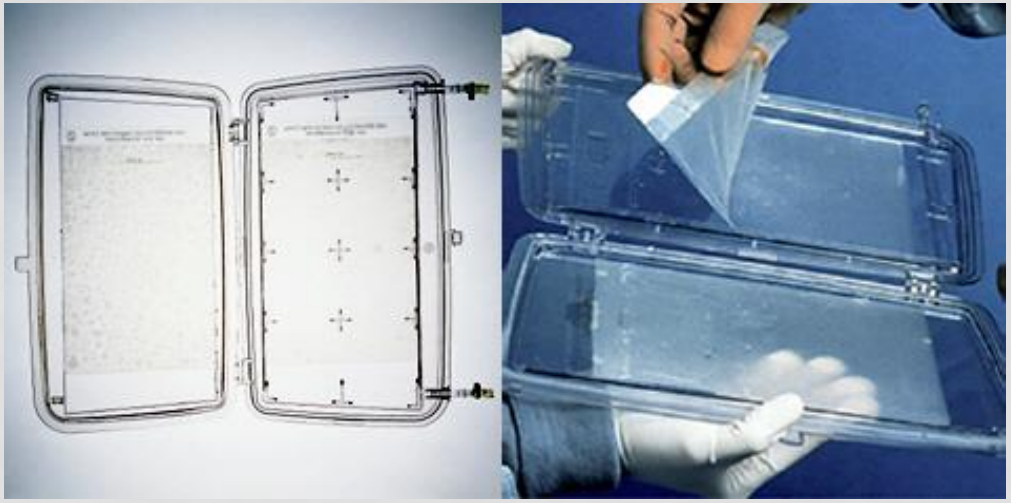
BIOLOGIC DRESSINGS

Human allograft (cadaver skin)

Porcine xenograft

Synthetic bilaminates

- ✓ Cryopreserved human fibroblast-derived dermal substitutes
  - ➢ Transcyte
  - ➢ Dermagraft
- ✓ Silicone/collagen
  - ➢ Biobrane



BIOBRANE, SUPRATHEL

Artificial skin for temporary coverage
For partial thickness burns
Leave on until wound has healed
Decreases the need for dressing changes

Goals

- ✓ **Ease of use**
- ✓ **Effective**
  - ➢ **Heat loss prevention**
  - ➢ **Coverage to prevent contamination**
  - ➢ **Antimicrobial coverage**
- ✓ **Cost**

BURN SURGERY

Ideally deep burns should be excided within 72 hours post injury

- Early excision
- Better esthetic and functional outcome
- Shortens hospital stay
- Decreases infection, sepsis, mof etc
- Saves money!

TANGENTIAL EXCISION

Remove tissue tangentially until viable tissue

Bleeding may be a problem

30-40% can be excised during one operation

Better esthetic outcome

FASCIAL EXCISION

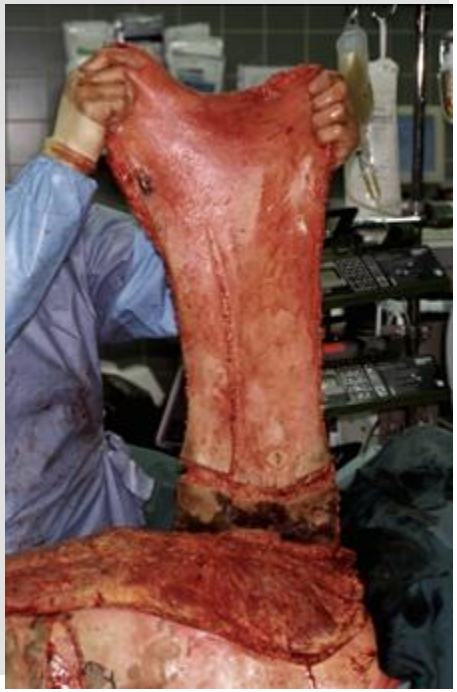
In life threatening and very big/deep burns

Remove burn tissue on the level of the fascia

Fast, bleeds less than tangential excision

Up to 80% TBSA can be removed in one operation

Poor esthetic outcome



ACUTE BURN CARE

SUMMARY

Anticipate need for early intubation

Know when to suspect

- CO poisoning
- Inhalation injury
- 100% Oxygen – good thing

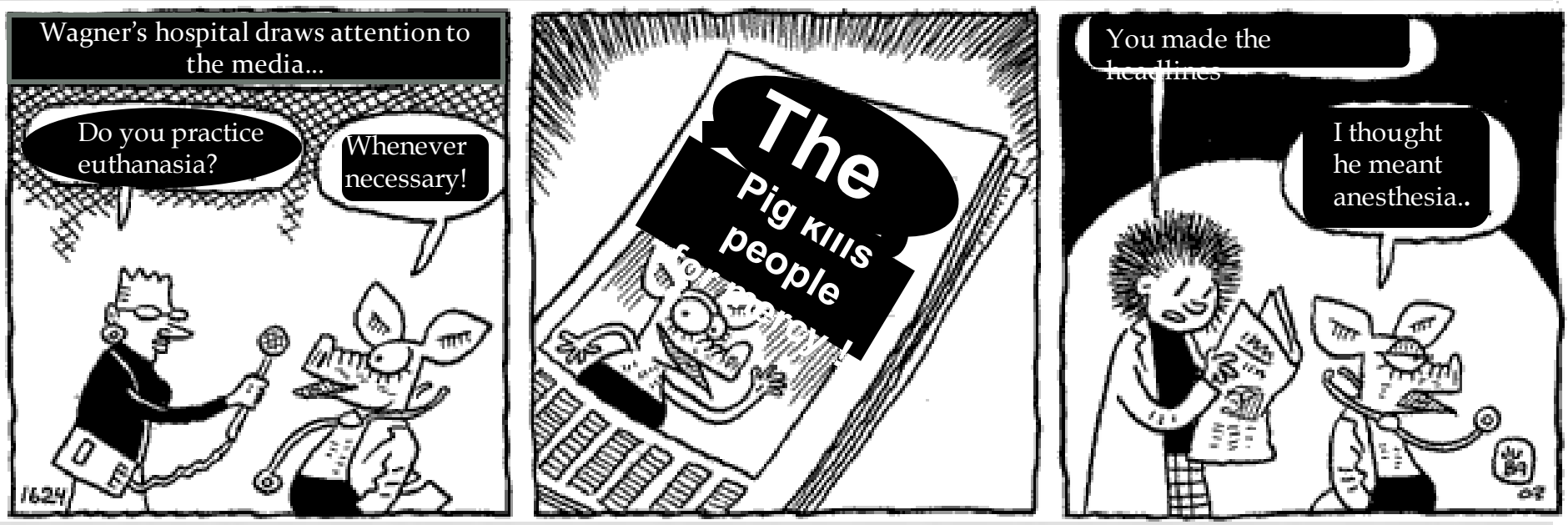
Bronchoscopy best way to diagnose inhalation injury

ACUTE BURN CARE

SUMMARY

- Calculate fluid requirements then watch clinical response
- Tetanus immunization – Good thing
- Prophylactic antibiotics – Bad thing
- Remember:
  - Escharotomy
  - Fasciotomy

Wagner


Wagner's hospital draws attention to the media...
Do you practice euthanasia?
Whenever necessary!
1624
The
Pig kills people
You made the headlines
I thought he meant anesthesia..

SPECIAL BURNS

ELECTRICAL INJURY

Arrythmias common, including VT, VF

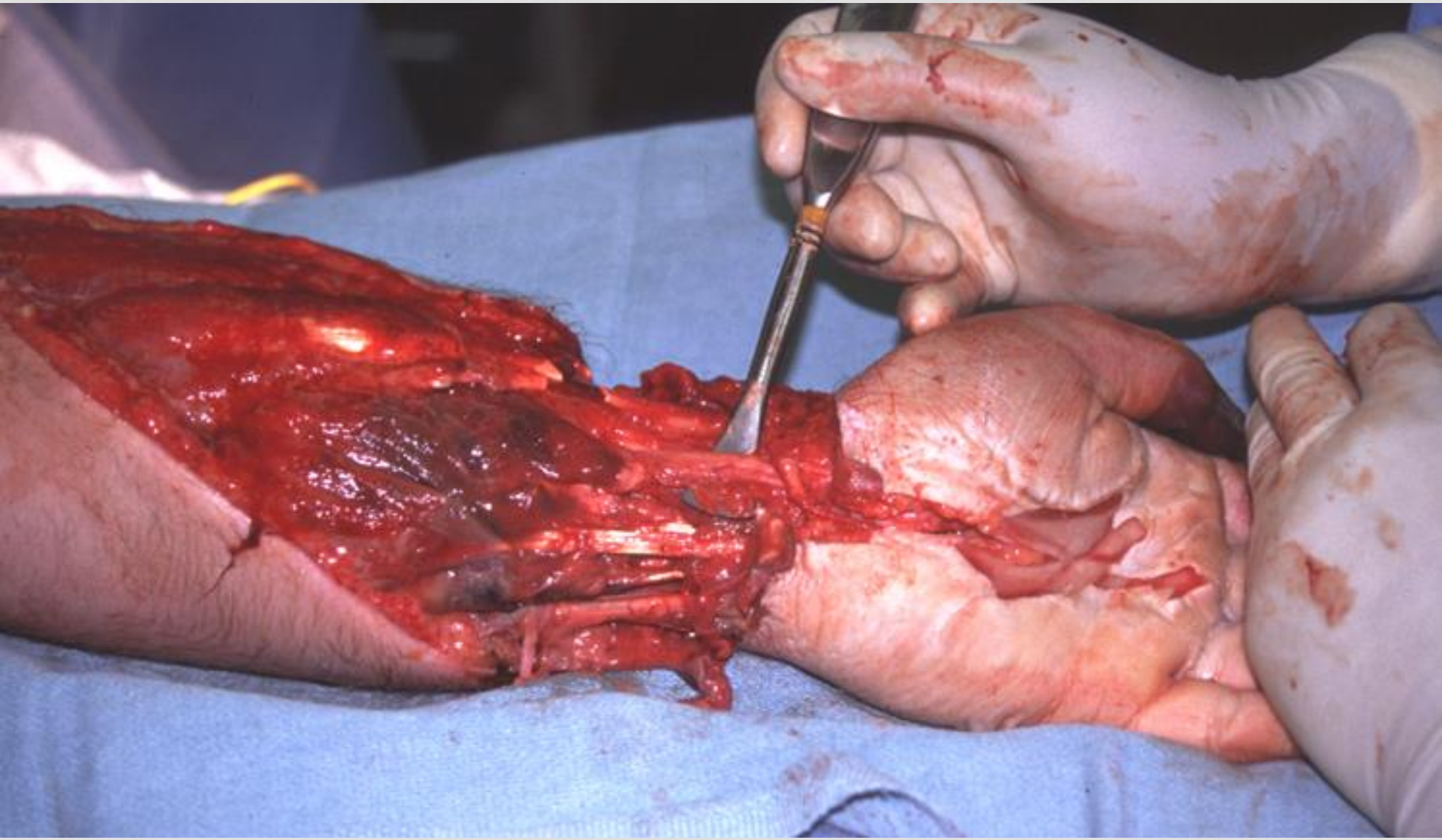
- ✓ ACLS

48 hr EKG monitoring if:

- ➢ LOC
- ➢ Abnormal EKG

Associated fractures common

- ✓ falls
- ✓ tetanic contraction
- ✓ pinal injuries



ELECTRICAL INJURY II

Vascular injury must be excluded:

- Electrical current may damage intima
- Arteriography may be necessary
  - Identify level of injury
  - Extent of damage for amputation
- Fasciotomy indicated when
  - Hard, stony muscle
  - Loss of pulses
  - Compartment pressure > 30mmHg

Whenever you think the patient may need a fasciotomy, prepare to done one

ELECTRICAL INJURY III

Renal injury secondary to hemochromagens (Hgb, Mgb)

- Increase IV fluids
- Follow CK
- Consider Mannitol 25g bolus (300 g i
  24 hrs)
- Consider $NaHCO_3$: keep urine pH >
  (alkalization of urine)

ELECTRICAL INJURY IV

Neurologic Injury

Medullary centers:

- ✓ respiration often affected in high voltage injury, lightning

Spinal cord damage common

Early deficits may be transient

Late deficits are usually permament

Cataracts may appear weeks to months after head and neck injury

WHITE PHOSPHORUS

Used in grenades, flares, mines, shells, bombs

- Burns on contact with air
- Wound may smoke or burn
- Irrigate copiously with saline
- 0.5 – 1% Copper sulfate wash followed by copious irrigation turns particles blue-gray, impedes ignition, allows ID of particles

HYDROFLOURIC ACID (HF)

Hydroflouric acid (HF)

- ✓ Used for glass etching, metal cleaning
  - ➢ Naval shipyards
- ✓ Prolonged irrigation with Benzalkonium chloride
- ✓ Topical application of Calcium gluconate gel
- ✓ Local injection of 10% Calcium gluconate solution for severe pain
- ✓ Treat hypocalcemia as necessary

OTHER AGENTS

Tar, Bitumen:

- ✓ Sticks to skin
- ✓ Hot material cooled with cold water
- ✓ Do not use petroleum solvents, instead use Vaseline dressing changes

Phenol

- ✓ Flush copiously with water
- ✓ Use glycerol, polyethylene glycol, or propylene glycol to remove residual phenol